

Operational Diagnostics Report

The Nairobi Bottleneck Investigation

Oil & Gas Pod | Week 5 Stakeholder Defense

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Executive Summary

Nairobi depot's throughput fell by 34.4% starting 01 February 2026 — from an average of 1,052.5 barrels per shift to 690.6 barrels per shift — while the other three depots (Mombasa, Kisumu, Eldoret) stayed essentially flat over the same window (within 1% either way).

The drop traces to a single machine: pump NBI-P03. It is the only Nairobi machine carrying missed-maintenance incidents (529 of its 566 recorded shifts, 93.5%), while the depot's other three pumps show zero missed-maintenance incidents and normal throughput throughout the year.

On affected shifts, NBI-P03 runs at a 40.6% throughput loss relative to its own healthy baseline, and it accounts for the entire quantified throughput loss at the depot — clearing the classic Pareto 80/20 threshold on its own. Ambient temperature and SCADA voltage — the two alternative explanations available in the dataset — both show negligible correlation with throughput (temperature: $r = 0.17$; voltage: $r = -0.03$) and are ruled out as drivers.

Recommendation: prioritize an immediate maintenance intervention on NBI-P03. Given the size and concentration of the loss, this single fix should recover the large majority of Nairobi's throughput gap without requiring changes at any other depot.

1. Dataset & Methodology

The analysis uses `Mystery_Ops.csv`: 2,920 daily depot-shift operational records for 2026, spanning four depots (Nairobi, Mombasa, Kisumu, Eldoret), balanced at 365 days x 2 shifts per depot. The primary KPI is `throughput_barrels`.

1.1 Data quality

Three fields have missing values: `operator_id` (18 records), `throughput_barrels` (5 records, a simulated export gap), and `temperature_c` (20 records, concentrated in a known Nairobi sensor outage). None of these gaps coincide with, or explain, the throughput break investigated below — they are noted here as a data-quality flag for the operations team, separate from the root-cause finding.

1.2 Analytical approach

- Univariate & bivariate profiling of the KPI and candidate drivers
- Anomaly detection via within-depot IQR flagging, cross-checked against a time-series view
- Root-cause drill-down by time period and by machine
- Pareto analysis to quantify each machine's share of total lost throughput
- Correlation analysis to test — and rule out — temperature and voltage as confounding drivers

Full code, narration, and all chart outputs are in the companion notebook, `week5_diagnostics_analysis.ipynb`.

2. Data Profiling & Distributions

The distribution of `throughput_barrels` across all depots is visibly bimodal (Figure 1) — a strong early signal that two distinct operating regimes are mixed in the data, rather than one population with random noise. Depot and shift record counts are perfectly balanced, ruling out sampling bias; `incident_type` is dominated by "none" with "missed_maintenance" as a distant second, already pointing toward where the investigation should focus.

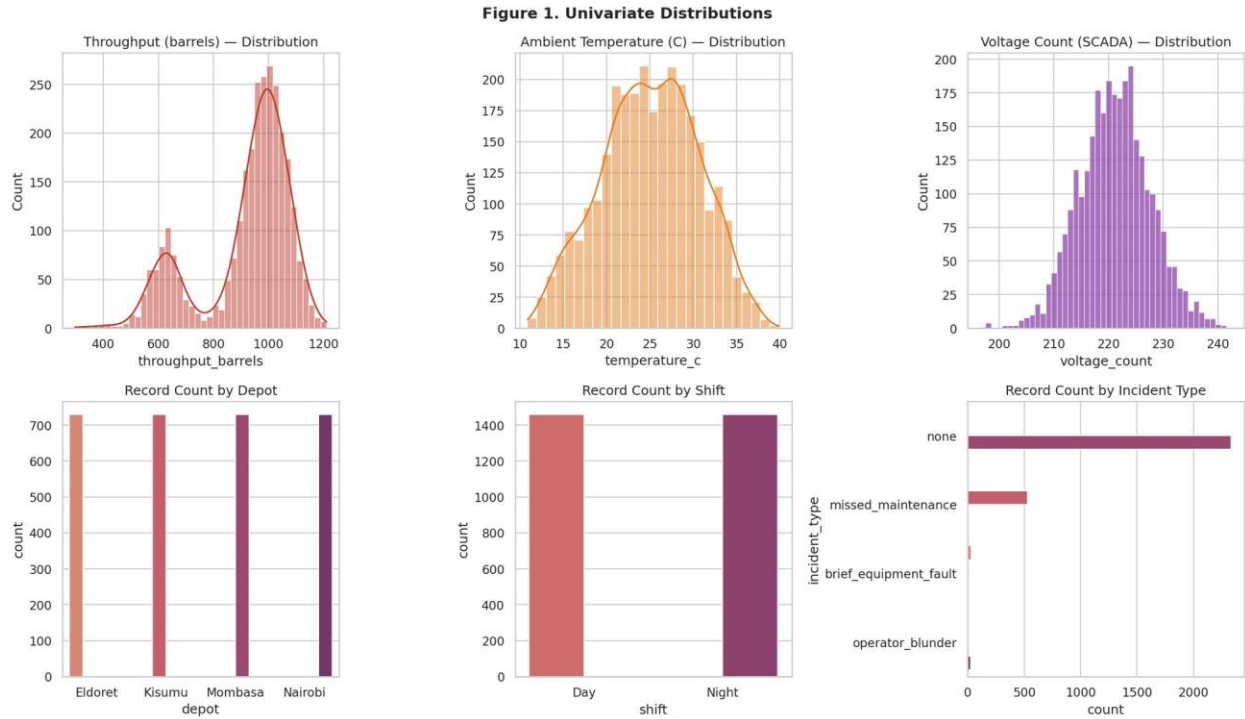


Figure 1. Univariate distributions of throughput, temperature, voltage, and category counts.

2.1 Throughput by depot and shift

Breaking throughput down by depot (Figure 2) shows Nairobi's entire distribution sitting below the other three depots, which look similar to one another. Shift (Day vs Night) shows no meaningful difference, ruling shift out as a driver.

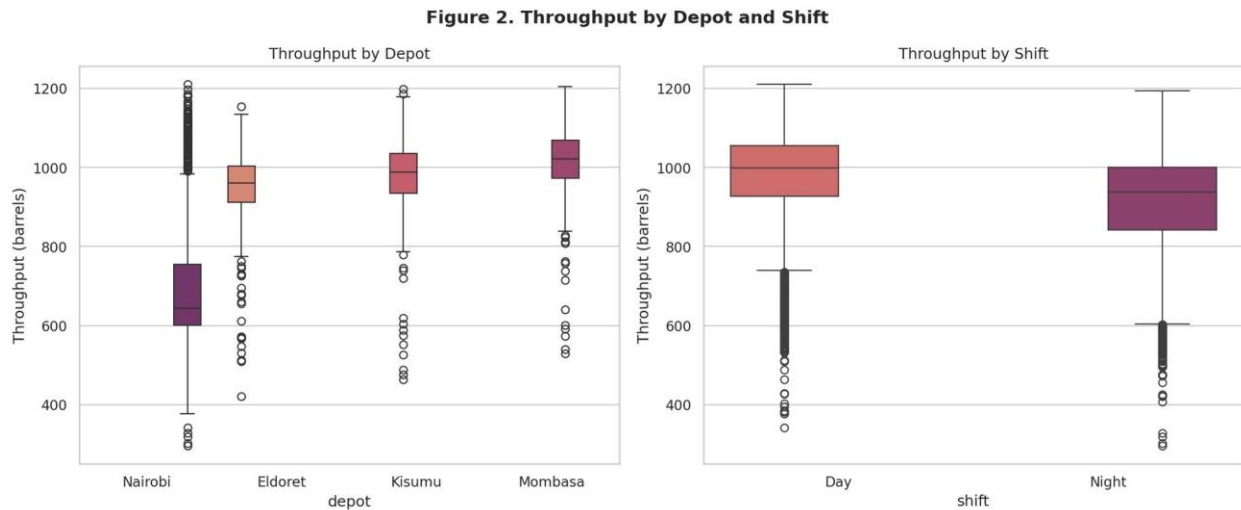


Figure 2. Throughput distribution by depot (left) and by shift (right).

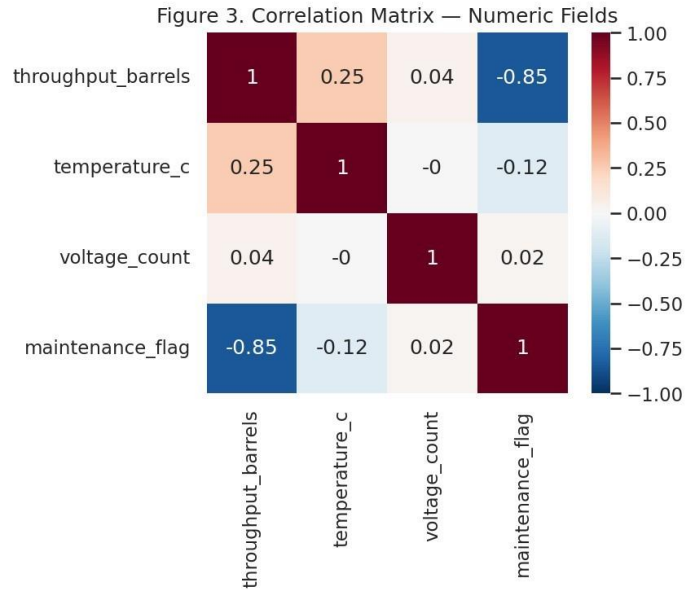


Figure 3. Correlation matrix of numeric fields — `maintenance_flag` stands out; `temperature` and `voltage` do not.

3. Anomaly Detection

Two significant anomalies were identified using within-depot IQR flagging cross-checked against a time-series view (Figure 4):

- A structural break in Nairobi's throughput starting 01 February 2026 — a step-down across the whole cloud of points, not a handful of isolated dots. This break is large enough that it distorts naive IQR flagging: because post-break low readings now form the majority of Nairobi's data, a within-depot IQR rule actually flags the pre-break high readings as the "outliers." That inversion is itself informative — static outlier rules can miss or misread a regime change, which is why the investigation moves to an explicit time-based drilldown next.
- A materially wider spread of low-side outliers in Nairobi than in any other depot, visible in the box plot's long lower whisker — consistent with a recurring equipment problem rather than isolated equipment faults, which are spread thinly across all four depots.

Figure 4. Anomaly Detection

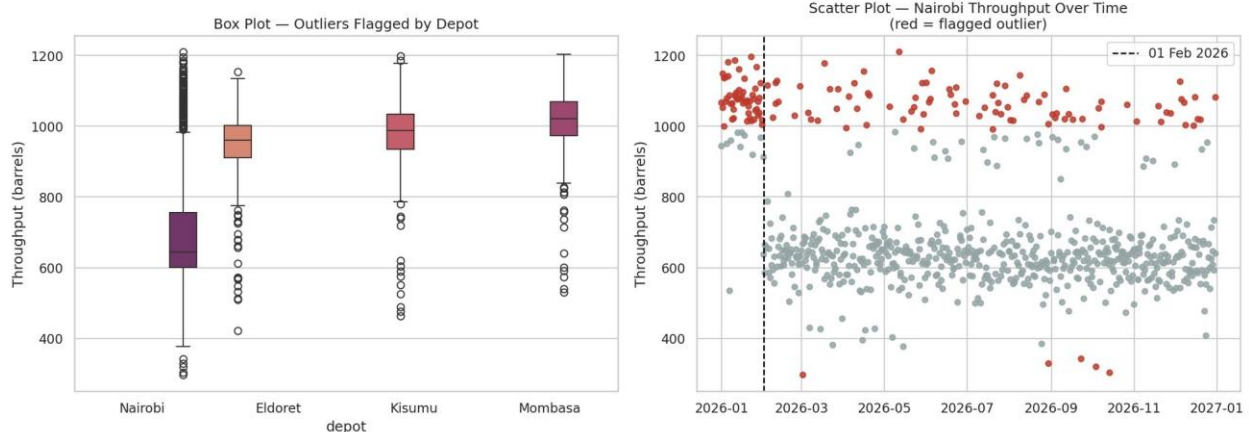


Figure 4. Box plot of throughput by depot (left) and Nairobi throughput over time (right), with the 01-Feb-2026 break marked.

4. Root Cause Analysis

4.1 Time-based drill-down

Comparing mean throughput before and after 01 February 2026 by depot confirms the break is Nairobi-specific and time-specific, not a market-wide seasonal effect:

Depot	Before 01-Feb (avg)	From 01-Feb (avg)	% Change
Eldoret	957.1	951.1	-0.6%
Kisumu	980.8	979.1	-0.2%
Mombasa	1,025.1	1,015.0	-1.0%
Nairobi	1,052.5	690.6	-34.4%

4.2 Machine-level drill-down

Within Nairobi, throughput and incident history split cleanly by machine:

Machine	Records	Mean Throughput	Missed-Maint. Incidents	% of Records
NBI-P01	66	1,021.4	0	0.0%
NBI-P02	51	1,026.1	0	0.0%
NBI-P03	566	633.1	529	93.5%
NBI-P04	47	1,031.3	0	0.0%

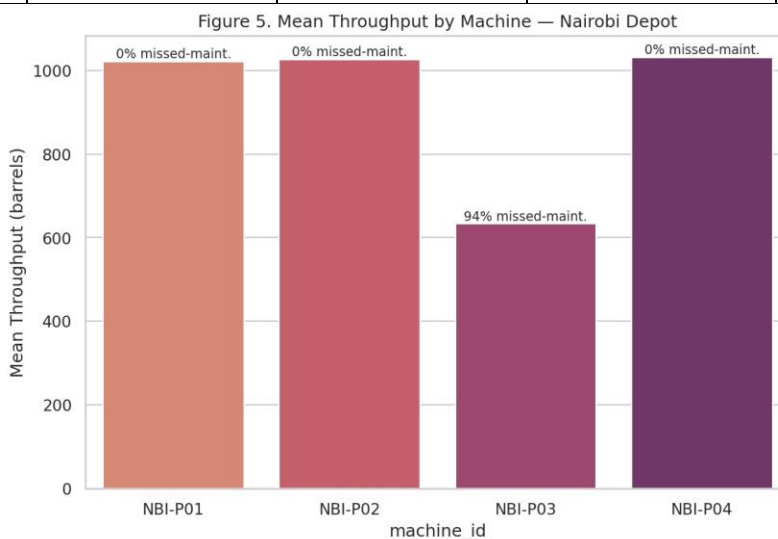


Figure 5. Mean throughput by machine, Nairobi depot — NBI-P03 is the clear outlier.

The time-based and machine-based drill-downs converge on the same answer independently: pump NBI-P03 is the root cause. The other three Nairobi machines behave like normal, healthy equipment throughout the year.

5. Pareto Analysis — Quantifying the Impact

Comparing NBI-P03's mean throughput on missed-maintenance shifts against its own healthy baseline (mean throughput on incident-free shifts) quantifies the loss directly in barrels:

Machine	Healthy Baseline	Missed-Maint. Mean	% Loss / Shift	Affected Shifts	Total Barrels Lost	Cumulative %
NBI-P03	1,052.3	625.1	40.6%	529	226,027	100.0%

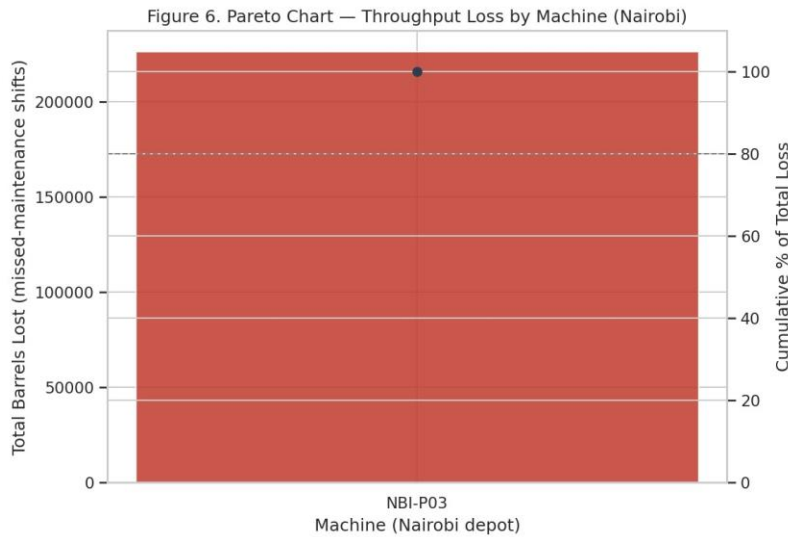


Figure 6. Pareto chart of throughput loss by machine — NBI-P03 accounts for 100% of the quantified loss.

Because NBI-P03 is the only Nairobi machine with any missed-maintenance incidents, it accounts for 100% of the depot's quantified throughput loss — a single-cause Pareto result well beyond the standard 80/20 threshold. Per affected shift, the machine loses 40.6% of its healthy-baseline output, consistent with progressive maintenance-driven degradation rather than random noise.

6. Ruling Out Weather and Voltage

Before attributing the loss to maintenance, the two alternative explanations available in the dataset were tested directly against Nairobi throughput:

Variable	Correlation with Throughput	Interpretation
temperature_c	0.17	Negligible — not a driver
voltage_count	-0.03	Negligible — not a driver

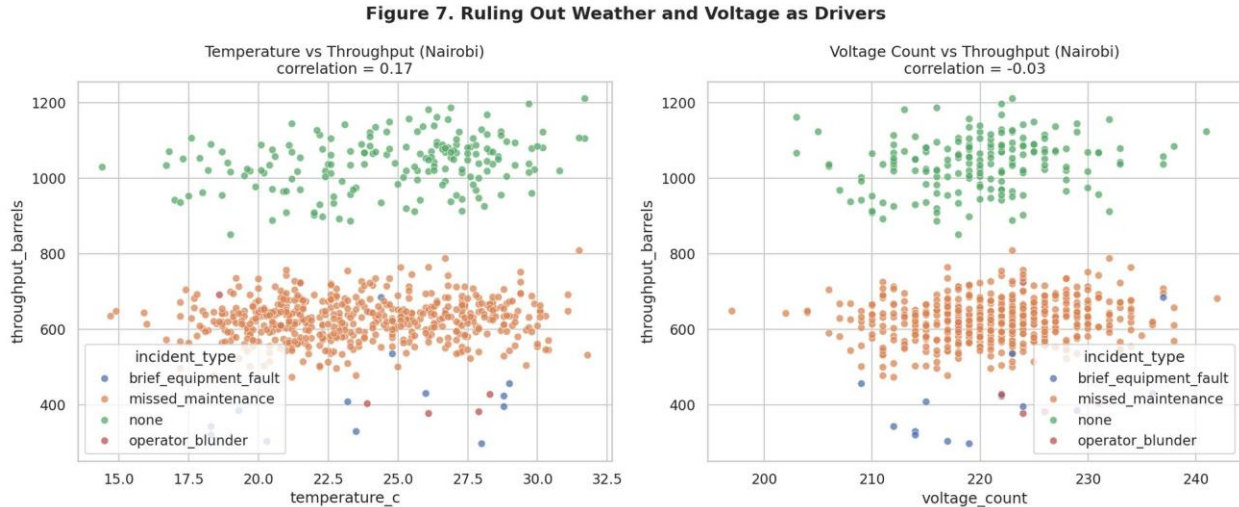


Figure 7. Temperature and voltage vs. throughput, colored by incident type — the split tracks incident type, not the x-axis variable.

In both scatter plots, the low-throughput points are colored almost entirely by missed_maintenance and are distributed across the full range of temperature and voltage — the drop tracks the incident type, not the environmental reading. This rules out weather and voltage as drivers and reinforces the maintenance-based root cause identified above.

7. Conclusions & Recommendation

- What: Nairobi depot's throughput fell 34.4% starting 01 February 2026 — a structural break, not random noise.
- Where: the drop is concentrated entirely in one machine, NBI-P03, surrounded by three healthy Nairobi pumps.
- Why: NBI-P03's loss coincides with missed-maintenance incidents on 93.5% of its shifts; it accounts for 100% of Nairobi's quantified throughput loss (Pareto), while temperature and voltage — the available alternative explanations — both show negligible correlation with throughput.

Recommendation: prioritize an immediate maintenance intervention on NBI-P03. Given the size and concentration of the loss, this single fix should recover the large majority of Nairobi's throughput gap and close most of the depot-level performance difference identified in Section 2 — without needing to touch the other three depots, which show only normal operational variation.